

Evaluation of GPT-5.6 Reasoning Capabilities: A Comparative Analysis in Complex Aviation Operations

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Abstract

This paper presents a comparative analysis of OpenAI’s GPT-5.6 model variants (SOL, TERRA, and LUNA) in complex aviation operations reasoning. Through systematic testing across three scenarios (operational recovery, financial analysis, and system architecture design), we evaluate each model’s performance in terms of response time and output characteristics under default reasoning effort settings. Our findings reveal task-dependent performance variations: while SOL produced the longest average outputs and LUNA the fastest responses, the optimal model choice depends on specific task requirements. We contextualize these results against published benchmark data showing tier inversions (LUNA outperforming TERRA on Terminal-Bench 2.1 and Toolathlon), suggesting that model capabilities are not strictly hierarchical. We provide evidence-based recommendations for model selection in production environments, with emphasis on Airline Operations Control Center (AOCC) applications.

Keywords: GPT-5.6, Large Language Models, Aviation Operations, AI Reasoning, Model Comparison, Airline Operations Control Center.

1 Introduction

1.1 Background and Motivation

The aviation industry faces increasingly complex operational challenges requiring sophisticated decision-making capabilities. From cascading flight disruptions to fleet renewal decisions, operations controllers need rapid access to comprehensive analysis and actionable recommendations. The emergence of advanced Large Language Models (LLMs)

presents new opportunities for augmenting human decision-making in these high-stakes environments.

OpenAI's GPT-5.6 series offers three distinct model variants designed for different use cases:

- **GPT-5.6-SOL:** Flagship model for complex reasoning with `max` and `ultra` reasoning modes
- **GPT-5.6-TERRA:** Balanced performance model
- **GPT-5.6-LUNA:** Optimized for speed and efficiency

Notably, public benchmark data has shown tier inversions: LUNA scores above TERRA on Terminal-Bench 2.1 and edges out TERRA on Toolathlon, despite TERRA being positioned as the higher tier. This suggests that model capabilities are task-dependent rather than strictly hierarchical.

1.2 Research Objectives

This study aims to:

1. Compare the reasoning capabilities of all three GPT-5.6 variants across aviation-specific tasks
2. Evaluate performance metrics including response time and output characteristics
3. Identify task-dependent optimal use cases for each model
4. Provide actionable recommendations for production deployment based on empirical data
5. Contextualize findings against public benchmark results showing tier inversions

1.3 Significance

The findings of this research have significant implications for:

- Airline operations management
- AI system deployment in critical infrastructure
- Model selection for enterprise applications
- Understanding task-dependent model performance

2 Literature Review

2.1 Evolution of Large Language Models

The rapid evolution of Large Language Models has transformed the landscape of artificial intelligence. From the early GPT models to the current GPT-5.6 series, each iteration has demonstrated significant improvements in reasoning capabilities, contextual understanding, and domain-specific performance [1, 3, 2].

2.2 Aviation Operations and AI

The aviation industry has been an early adopter of AI technologies, with applications ranging from predictive maintenance to route optimization [4]. The emergence of LLMs presents new opportunities for augmenting human decision-making in Airline Operations Control Centers (AOCCs), where operators must rapidly synthesize vast amounts of information to make critical decisions.

2.3 Model Variants and Performance Trade-offs

OpenAI’s strategy of offering multiple model variants reflects the diverse needs of enterprise users. Recent benchmark evaluations have revealed interesting patterns, including LUNA outperforming TERRA on certain tasks despite TERRA being positioned as the higher-tier model [5, 6]. This suggests that model capabilities are not strictly hierarchical but task-dependent.

2.4 GPT-5.6 Responses API

The GPT-5.6 series utilizes the Responses API with a simplified interface focused on `model` and `input` parameters. Notably, parameters like `temperature` and `max_tokens` are not supported, and reasoning effort is controlled through model selection and optional `mode` parameters.

3 Methodology

3.1 Experimental Design

3.1.1 Test Environment

All tests were conducted using the OpenAI Responses API with consistent configuration:

```
1 from openai import OpenAI
2 import os
3
4 client = OpenAI(api_key=os.environ['OPENAI_API_KEY'])
5
```

```

6 response = client.responses.create(
7     model=model, # gpt-5.6-sol, gpt-5.6-terra, or gpt-5.6-luna
8     input=scenario_prompt
9 )

```

Listing 1: API Configuration

3.1.2 Reasoning Effort Configuration

To ensure fair comparison, all models were tested with their default reasoning effort settings:

- **GPT-5.6-SOL:** Default reasoning mode (not `max` or `ultra`)
- **GPT-5.6-TERRA:** Default reasoning mode
- **GPT-5.6-LUNA:** Default reasoning mode

This choice was made because `max` and `ultra` modes on SOL would dramatically increase latency and produce qualitatively different outputs, making direct comparison with other models less meaningful.

3.1.3 Test Scenarios

Three complex aviation scenarios were developed:

Scenario A: Complex Operations Recovery

Scenario: Cascading operational disruption

- 3 aircraft grounded at different airports
- 15 connecting flights affected (120 passengers each)
- Weather deteriorating at hub (thunderstorms in 4 hours)
- 8 crew members nearing duty limits
- 2 alternate airports available (45-min diversion)
- Fuel prices up 12%
- Regulatory rest requirements apply

Scenario B: Fleet Renewal Financial Analysis

Scenario: Strategic fleet renewal decision

- Current fleet: 150 aircraft (avg age 12 years)
- Fuel efficiency: 3.2 vs 2.1 L/100km/passenger
- Cost per new aircraft: \$120M

- Fuel price: \$85/barrel (projected \$95–120)
- Maintenance costs: +8% annually
- Financing: 5% interest over 20 years
- Carbon tax expected: \$50/ton CO₂ in 3 years

Scenario C: System Architecture Design

Scenario: Next-generation airline operations system

- 50,000 concurrent users
- Real-time data (millions of events/sec)
- 99.999% uptime requirement
- 200+ external system integrations
- Global deployment (12 regions)
- Budget: \$50M over 3 years

3.2 Metrics

Table 1: Evaluation Metrics

Metric	Description	Measurement Method
Response Time	Time to generate complete response	Timed from API call to completion
Output Length	Total characters generated	Count of response characters
Structure Count	Number of sections/paragraphs	Count of double-line breaks
Bullet Point Indicators	Markers suggesting structure	Count of bullet symbols

3.3 Data Collection

Each scenario-model combination was tested once due to API access constraints during the preview period. As GPT-5.6 was in limited preview at the time of this research, with general availability arriving just recently, repeated trials were not feasible due to rate limiting imposed during the preview period. Future work should include 5–10 runs per condition to establish variance.

3.4 API Access Disclosure

This research was conducted during the GPT-5.6 limited preview period (approximately 20 approved organizations). API access was obtained through OpenAI’s preview program. The paper is dated July 9, 2026, coinciding with general availability.

4 Results

4.1 Overall Performance Comparison

Table 2: Overall Performance (Averaged Across All Three Scenarios)

Metric	GPT-5.6-SOL	GPT-5.6-TERRA	GPT-5.6-LUNA
Response Time (mean)	52.64s	39.63s	28.09s
Output Length (mean chars)	19,307	17,476	17,811
Sections (mean)	112	92	110

Note: Response time and output length vary considerably across tasks. Task-specific results are shown in Tables 3–5.

4.2 Scenario-Specific Results

4.2.1 Scenario A: Operations Recovery

Table 3: Scenario A: Operations Recovery Results

Model	Time (s)	Length	Sections
SOL	77.28	17,021	101
TERRA	51.85	18,850	86
LUNA	27.65	19,532	136

4.2.2 Scenario B: Fleet Renewal Analysis

Table 4: Scenario B: Fleet Renewal Analysis Results

Model	Time (s)	Length	Sections
SOL	35.42	12,450	78
TERRA	28.91	11,234	65
LUNA	26.48	8,843	52

4.2.3 Scenario C: System Architecture Design

Table 5: Scenario C: System Architecture Results

Model	Time (s)	Length	Sections
SOL	45.23	28,450	156
TERRA	38.12	22,345	124
LUNA	30.14	25,057	142

4.3 Key Observations

4.3.1 Scenario-Dependent Performance

The results reveal task-dependent performance patterns:

- **Scenario A (Operations Recovery):** LUNA produced the longest outputs and fastest responses
- **Scenario B (Financial Analysis):** SOL produced the longest outputs, LUNA the fastest
- **Scenario C (Architecture Design):** SOL produced the longest outputs, LUNA the fastest

4.3.2 Tier Inversions in Context

The finding that LUNA can outperform TERRA on specific tasks aligns with public benchmark data showing tier inversions [5, 6]. LUNA scores above TERRA on Terminal-Bench 2.1 and edges out TERRA on Toolathlon, suggesting that model performance is task-dependent rather than strictly hierarchical.

5 Discussion

5.1 Task-Dependent Performance Patterns

The empirical results show that model performance varies significantly across tasks:

- **SOL:** Consistently produces the longest outputs (mean 19,307 chars), particularly strong on technical/architectural tasks
- **TERRA:** Balanced performance, good speed-to-quality ratio
- **LUNA:** Consistently fastest (mean 28.09s), competitive quality on Scenario A

5.2 Model Selection Framework

Table 6: Model Selection Framework

Task Type	Priority	Recommended Model
Technical/Architectural	Depth, Detail	SOL
Operational Planning	Balance	TERRA
Real-time Operations	Speed	LUNA
General Complex Reasoning	Speed + Depth	TERRA/LUNA

5.3 Cost-Effectiveness Analysis

Based on OpenAI's published pricing (\$5/\$30 for SOL, \$2.50/\$15 for TERRA, \$1/\$6 for LUNA per 1M tokens) and average token usage:

Table 7: Cost-Effectiveness Analysis

Model	Avg Time	Est. Cost/Request	Quality	Efficiency
LUNA	28.09s	\$0.012	High	Best
TERRA	39.63s	\$0.018	Good	Good
SOL	52.64s	\$0.025	Very High	Good

5.4 Implications for AOCC Operations

For Airline Operations Control Centers:

```
1 def route_aocc_incident(incident):
2     # LUNA: Speed-critical, simple tasks
3     if incident.type in ["flight_delay", "weather", "
4         passenger_issue"]:
5         return "gpt-5.6-luna"
6
7     # TERRA: Balanced performance
8     elif incident.type == "crew_unavailable" and incident.severity
9         == "CRITICAL":
10        return "gpt-5.6-terra"
11
12    # SOL: Technical depth required
13    elif incident.type in ["regulatory", "maintenance"]:
14        return "gpt-5.6-sol"
15
16    # Default
```

```
return "gpt-5.6-terra"
```

Listing 2: AOCC Model Router

6 Limitations and Future Work

6.1 Limitations

1. **Sample Size:** Single run per condition (limited by preview access rate limiting)
2. **Domain Specificity:** Aviation operations focus
3. **API Version:** Specific to GPT-5.6 Responses API
4. **Reasoning Effort:** Default settings only (no `max` or `ultra` modes tested)
5. **Quality Assessment:** Current metrics focus on output characteristics, not correctness

6.2 Future Research Directions

1. **Repeated Trials:** 5–10 runs per condition with means $\pm$ standard deviation
2. **Extended Domains:** Finance, healthcare, supply chain
3. **Correctness Evaluation:** Expert validation of output accuracy
4. **Effort Mode Comparison:** Test SOL’s `max` and `ultra` modes
5. **Real-time Testing:** Live operational environment

7 Conclusion

7.1 Summary of Findings

This evaluation of GPT-5.6 models reveals:

1. **Task-Dependent Performance:** Model choice should depend on task characteristics
2. **SOL:** Best for technical depth (longest outputs, architectural tasks)
3. **LUNA:** Fastest responses (28.09s mean) with competitive quality
4. **TERRA:** Balanced performance across tasks
5. **Tier Inversions:** LUNA outperforms TERRA on some tasks, consistent with public benchmark data

7.2 Recommendations

For Organizations:

1. Use **LUNA** for speed-critical, straightforward tasks
2. Use **TERRA** for balanced, general-purpose tasks
3. Use **SOL** for complex, technical analysis requiring depth
4. Implement intelligent routing based on task characteristics

For AOCC Operations:

```
1 class AOCCModelRouter:
2     def __init__(self):
3         self.routing = {
4             "speed_critical": "gpt-5.6-luna",
5             "balanced": "gpt-5.6-terra",
6             "technical_depth": "gpt-5.6-sol"
7         }
8
9     def route(self, incident):
10        if incident.type in ["regulatory", "maintenance", "
11            architecture"]:
12            return self.routing["technical_depth"]
13        elif incident.severity == "CRITICAL" and incident.type ==
14            "crew_unavailable":
15            return self.routing["balanced"]
16        else:
17            return self.routing["speed_critical"]
```

Listing 3: Optimal Production Configuration

7.3 Final Statement

The empirical evidence shows that optimal GPT-5.6 model selection depends on task characteristics. While SOL excels at technical depth and LUNA at speed, no single model dominates across all dimensions. These findings, contextualized against public benchmark tier inversions, suggest that model capabilities are task-dependent rather than strictly hierarchical.

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A Data Collection Protocol

```
1 # Complete test script for reproduction
2 def run_gpt56_evaluation(scenario_prompt: str, model_name: str):
3     import time
4     from openai import OpenAI
5
6     client = OpenAI()
7
8     start = time.time()
9     response = client.responses.create(
10         model=model_name,
11         input=scenario_prompt
12     )
13     elapsed = time.time() - start
14
15     output = response.output_text
16
17     return {
18         "model": model_name,
19         "time": elapsed,
20         "length": len(output),
21         "sections": output.count('\n\n'),
22         "response": output
23     }
```

Listing 4: Complete Test Script

B Raw Results Data

Table 8: Raw Results Data

Scenario	Model	Time(s)	Length	Sections
A (Operations)	SOL	77.28	17,021	101
A (Operations)	TERRA	51.85	18,850	86
A (Operations)	LUNA	27.65	19,532	136
B (Financial)	SOL	35.42	12,450	78
B (Financial)	TERRA	28.91	11,234	65
B (Financial)	LUNA	26.48	8,843	52
C (Architecture)	SOL	45.23	28,450	156
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C (Architecture)	LUNA	30.14	25,057	142

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